Keel Mori Theorem

Thesis in partial fulfillment of the requirements for the degree
Master of Science (M.Sc.)

submitted by
M. Sc.
Jonathan Wassermann

Matriculation number
XXXXXX

Under the scientific supervision of
Dr. Dominic Oliver Brooks Bunett

Under the scientific guidance of
My advisor, M.Sc.

Berlin, March 2026

Technische Universität Berlin
Fakultät II – Mathematik und Naturwissenschaften
Institut für Mathematik
Algorithmische Algebra

Affidavit

Deutsch

1) Hiermit versichere ich, dass ich die vorliegende Arbeit ohne Hilfe Dritter und ausschließlich unter Verwendung der aufgeführten Quellen und Hilfsmittel angefertigt habe. Alle Stellen, die den benutzten Quellen und Hilfsmitteln unverändert oder inhaltlich entnommen sind, habe ich als solche kenntlich gemacht.
2) Sofern generative KI-Tools verwendet wurden, habe ich Produktnamen, Hersteller, die jeweils verwendete Softwareversion und die jeweiligen Einsatzzwecke (z.B. sprachliche Überprüfung und Verbesserung der Texte, systematische Recherche) benannt. Ich verantworte die Auswahl, die Übernahme und sämtliche Ergebnisse des von mir verwendeten KI-generierten Outputs vollumfänglich selbst.
3) Die Satzung zur Sicherung guter wissenschaftlicher Praxis an der TU Berlin vom 15. Februar 2023[1] habe ich zur Kenntnis genommen.
4) Ich erkläre weiterhin, dass ich die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegt habe.

M. Sc.
Jonathan Wassermann
Berlin, 28. März 2026

[1] https://www.static.tu.berlin/fileadmin/www/10002457/K3-AMBl/Amtsblatt_2023/Amtliches_Mitteilungsblatt_Nr._16_vom_30.05.2023.pdf

English

1) I hereby declare that I have written this thesis without the help of third parties and exclusively using the sources and aids listed. I have marked as such all passages that are taken from the sources and aids used, either unchanged or in terms of content.
2) Where generative AI tools were used, I have stated the product name, manufacturer, the software version used and the respective purposes (e.g. linguistic review and improvement of texts, systematic research). I am fully responsible for the selection, adoption and all results of the AI-generated output I used.
3) I have taken note of the Statutes for Safeguarding Good Scientific Practice at TU Berlin dated February 15, 2023[2].
4) I further declare that I have not yet submitted the work in the same or a similar form to any other examination authority.

M. Sc.
Jonathan Wassermann
Berlin, March 28, 2026

[2]https://www.static.tu.berlin/fileadmin/www/10002457/K3-AMBl/Amtsblatt_2023/Amtliches_Mitteilungsblatt_Nr._16_vom_30.05.2023.pdf

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Masterarbeit
für <Herrn/Frau> <Vorname Nachname>
Matr.-Nr.: <123456>

Thema der Arbeit / Thesis topic (bitte klar benennen)

Text der Aufgabenstellung / Task description, including (**bitte ersetzen**)
- What is the challenge? What is the goal of the thesis?
- How shall this goal be achieved in terms of methods, models, experiments, etc.?

Bei der Bearbeitung der Aufgabenstellung sind die fachgebietsinternen Regeln für den Umgang mit generativen KI-Tools[1] zu beachten.

Prof. Dr.-Ing. habil. Jens-Uwe Repke

d|b|t|a

[1] https://www.static.tu.berlin/fileadmin/www/10002163/01_Lehre/Fachgebietsinterne_Vorgaben_zur_Nutzung_von_generativer_KI_ENG.pdf

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Acknowledgements

Ein paar nette Worte / Some nice words...

Abstract

Deutsch

Dies hier ist ein Blindtext zum Testen von Textausgaben. Wer diesen Text liest, ist selbst schuld. Der Text gibt lediglich den Grauwert der Schrift an. Ist das wirklich so? Ist es gleichgültig, ob ich schreibe: „Dies ist ein Blindtext“ oder „Huardest gefburn“? Kjift – mitnichten! Ein Blindtext bietet mir wichtige Informationen. An ihm messe ich die Lesbarkeit einer Schrift, ihre Anmutung, wie harmonisch die Figuren zueinander stehen und prüfe, wie breit oder schmal sie läuft. Ein Blindtext sollte möglichst viele verschiedene Buchstaben enthalten und in der Originalsprache gesetzt sein. Er muss keinen Sinn ergeben, sollte aber lesbar sein. Fremdsprachige Texte wie „Lorem ipsum“ dienen nicht dem eigentlichen Zweck, da sie eine falsche Anmutung vermitteln. Dies hier ist ein Blindtext zum Testen von Textausgaben. Wer diesen Text liest, ist selbst schuld. Der Text gibt lediglich den Grauwert der Schrift an. Ist das wirklich so? Ist es gleichgültig, ob ich schreibe: „Dies ist ein Blindtext“ oder „Huardest gefburn“? Kjift – mitnichten! Ein Blindtext bietet mir wichtige Informationen. An ihm messe ich die Lesbarkeit einer Schrift, ihre Anmutung, wie harmonisch die Figuren zueinander stehen und prüfe, wie breit oder schmal sie läuft. Ein Blindtext sollte möglichst viele verschiedene Buchstaben enthalten und in der Originalsprache gesetzt sein. Er muss keinen Sinn ergeben, sollte aber lesbar sein. Fremdsprachige Texte wie „Lorem ipsum“ dienen nicht dem eigentlichen Zweck, da sie eine falsche Anmutung vermitteln.

Schlüsselwörter: *Schlüsselwort1, Schlüsselwort2, Schlüsselwort3*

English

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Keywords: *Keyword1, Keyword2, Keyword3*

Contents

List of Figures

List of Tables

List of Algorithms

List of Codes

List of Abbreviations

If the horizontal space is too small or too large for you abbreviations, change the allocated space by entering the longest abbreviation in the `\settowidth{}` command in `f_Abbreviations.tex`.

Numerics

ab	**a**ctive **b**ound
DAE	**D**ifferential-**a**lgebraic **e**quation (system)

Software

SUNDIALS	**Su**ite of **n**onlinear and **di**fferential-**al**gebraic equation **s**olvers

Alternatively, you can only have one list of abbreviations without further classification:

ab	**a**ctive **b**ound
DAE	**D**ifferential-**a**lgebraic **e**quation (system)
SUNDIALS	**Su**ite of **n**onlinear and **di**fferential-**al**gebraic equation **s**olvers

List of ToDos

1 Guidelines

This chapter introduces the guidelines for writing a thesis at the Process Dynamics and Operations Group. It is recommended to use LaTeX as the code of this style guide can directly be used for the thesis. However, a Word version of this template is also available. In the template, the correct fonts, font sizes, citation style, and so forth are already set.

1.1 Most Important Style Specifications for LaTeX and Word

- page format: A4, double page, justification, 11 pt for standard font size;
- line spacing:
  - Word: 1.2;
  - LaTeX: linespacing is set with the `setspace` package any may not be changed;
- fonts :
  - Word: Palatino Linotype (text) and Arial (headings);
  - LaTeX: the font types are specified within this template and may not be changed;
- margins:
  - Word: 38 mm (top), 45 mm (bottom), 20 mm (inside), 35 mm (outside); binding correction: 20 mm (only possible when double page is selected); exception: front page;

- ◦ LaTeX: the given settings for BCOR and DIV may not be changed;
- – maximum number of pages (only content, excluding lists of ... and appendix): 80 pages (bachelor thesis), 100 pages (master thesis);
- – the layout of the front page is fixed and must not be changed, neither in LaTeX nor in Word;
- – of course, loading additional packages in LaTeX for functionality is fine.

Attention: If these specifications are (partially) ignored, it will have an impact on the evaluation.

1.2 First Steps

1) **carefully read this whole chapter.** It might look like there is a lot to read but you will get additional information on how to use the template, what additional software is out there (especially for drawing figures), and how to avoid issues with the template.

2) install LaTeX. On Windows, MikTeX[1] or TeXLive[2] are suitable distributions. MacTeX[3] or TeXShop[4] can be used on a Mac. On Linux, you typically also install TeXLive[5]; make sure to install an up-to-date version of TeXLive on a Linux machine. This template expects a TeXLive distribution $\geq$ 2019. It is not tested with older versions anymore.

3) install a suitable LaTeX editor. We recommend Texmaker[6], which is available for all operating systems. An extensive list of editors is available on Wikipedia[7].

[1] https://miktex.org/, February 2019
[2] https://tug.org/texlive/windows.html, February 2019
[3] http://www.tug.org/mactex/, February 2019
[4] https://pages.uoregon.edu/koch/texshop/, February 2019
[5] https://tug.org/texlive/quickinstall.html, February 2019
[6] http://www.xm1math.net/texmaker/, February 2019
[7] https://en.wikipedia.org/wiki/Comparison_of_TeX_editors, February 2019

4) install a suitable program for your literature. We recommend Jabref[8]. It is a Java-based, platform-independent program that generates appropriate `.bib` files for LaTeX. A short introduction to Jabref is given in Sec. 1.10.7.

5) set the language of the document in the `a_Packages.tex` file with the `babel` package. You will notice that this text is written in English while certain headings are in German. This is because the `babel` package is currently loaded with `ngerman` as default language (see Sec. 1.7).

6) change the necessary entries in `b_Meta.tex` (e.g., your name, matriculation number, etc.), set the right value of the `isDiss` variable and the `isMT` variable (if it is a thesis) in this file

7) check whether `biber` and `makeindex` were set up correctly. Detailed information on this matter can be found in Sec. 1.8.1

8) check whether you can compile this document without errors (see Sec. 1.8.2). This should always be the case as long as all necessary packages are installed. This template was successfully compiled with the TeXLive version given in the ReadMe of the git project (older LaTeX distributions potentially contain older package version and are thus not supported anymore). **Some problems appeared when users did not have the newest versions of the used packages in this template. If you run into trouble, please update all your packages.[9]** See some more instructions for TeXLive and MiKTeX in Sec. 1.8.3

9) remove the "Guidelines" chapter from this document by deleting it and removing it from the `0_Text.tex` file in the folder 03_Content

10) add your own `.bib` file for your references or use the present one (it is recommended to simply use the given one)

11) start writing your thesis – good luck!

[8] http://www.jabref.org/, February 2019
[9] https://tex.stackexchange.com/questions/55437/how-do-i-update-my-tex-distribution, January 2019

12) note that not all of the shown items in the following sections *must* be part of your thesis. If a certain aspect does not apply to you, for example a List of Algorithms, just remove it from the thesis.

1.3 General Information

1) the current “Prüfungsordnung” overrides the following rules if they contradict the “Prüfungsordnung”.

2) the thesis must be written in German or English.

3) a thesis is a scientific-technical documentation that must satisfy requirements regarding structure and form. It should be precisely formulated and well-written, i.e., no orthographic or grammar mistakes, etc.

4) the thesis should be logically structured.

5) the thesis should present its scientific-technical content while remaining comprehensible. Germans tend to formulate complex phrases with many sub-clauses. This should be avoided. Hence, the author should repeatedly put him- or herself into the position of the reader and evaluate the thesis in this regard.

6) the Figure, i.e., picture, diagram, photo, is preferred to long explanations.

7) results must be tractable. Hence, the applied methods, assumptions, boundary conditions, experiments, and computer codes must be pointed out and explained in sufficient detail.

8) calculations should be documented. This is of course difficult for large models. In this case, the code should be attached to the printed or digital appendix.

9) the thesis should focus on the central themes and aspects. Other information should be referenced appropriately, but does not have to be repeated extensively.

1.4 Form

1) physical units must always be given and are preferably stated in SI units. Units must not be stated in brackets:

   – *WRONG:* Pressure P [Pa];

   – *RIGHT:* Pressure P in Pa.

   The only correct use of square brackets is shown here for the voltage:
   [U] = V, i.e., the unit of the voltage is Volt.

2) a List of Symbols and a List of Abbreviations must be included. This is done with the `nomencl` and the `acro` package in this template. In addition, symbols should be explained in the text after their first appearance. A List of Algorithms or other lists can be added if necessary.

3) figures, tables, and equations must be numerated and referenced in the text. This is automatically done using the `caption` package (see Sec. 1.10) and the `\autoref` command (Sec. 1.10.10). For example, a Figure is named Figure chapter.Num (Figure 2.1). The numeration is done automatically in this template. In addition, figures and tables must also be explained and discussed in the text.

4) figures should be chosen to support comprehension. In particular, the most important details and relevant labels must be *readable*.

5) extensive tables or figures that are repeatedly referenced in the text should be put in the appendix.

6) information or data not generated by the author must always be referenced. Citations/references are used to

   – document and justify one's own statements,

   – differ between one's own statements and those made by others,

   – help the reader to assess the origin of a statement

All information not generated by the author must be marked with a short reference, which is accompanied by the extensive reference in the bibliography. It is not important if this information appears directly or indirectly in the text. We either use the authoryear or the numerical short citation.
The most important rule is: The references must be complete and follow a consistent format. This is more important than following a specific citation style. If possible, the DOI/ISBN of an article/book should be part of the citation. This is also included automatically in this template. The commands and some examples are shown in Sec. 1.10.7.

1.5 Appearence

1) the format of the page numbering and the appendix chapters is already specified and may not be changed.

2) there is a maximum of four indenture levels (chapter, section, subsection, paragraph) in the text and a maximum of three levels in the table of contents. The subsubsection should not be used as four numbers are bad style for structuring a text.

3) important aspects can be emphasized with *italics*, **bold writing**, or using the *emphasize command* `\emph`. Underlining words should be avoided.

4) paragraphs should not start in the last two lines of a page ("Schusterjunge" or orphen) or end in the first two lines of a page ("Hurenkind" or widow). This is automatically achieved with the `nowidow` package in this template.

1.6 Template Structure

This template contains several features given in Tab. 1.1. The structure of the template is described in Tab. 1.2. Questions, problems, or additional feature requests can be posted on the gitlab webpage of this template.[10]

[10] https://git.tu-berlin.de/dbta/templates/Thesis_template/issues, February 2021

Tab. 1.1: Template features.

No.	Feature
1)	Valid for theses or dissertations
2)	Usable in German or English depending on the `babel` settings
3)	Ready for use in online editors, such as Overleaf (see Sec. 1.9.2)
4)	List of Algorithms using the `algorithm2e` package and KOMA script
5)	List of Codes using the `listings` package and KOMA script
6)	List of Symbols (Latin, Greek, etc.) using the `nomencl` package
7)	List of Abbreviations using the `acro` package
8)	List of References using the `biblatex` package and `biber`
10)	Index with the `imakeidx` package
11)	ToDos and missing figures using the `todonotes` package
12)	Extra features, such as PDF/A compatibility or automatic indenting of TEX code
13)	various templates for typical elements of a thesis (figure, table, ...)

Tab. 1.2: File structure of template. Folder names are printed bold.

File name (without extension)	Description
`main`	Must be compiled Can be chosen as master document
00_Arara_and_Latexindent	
`localSettings`	local settings for automatic indenting More information in Sec. 1.9.3 and **??**
01_Document_administration	
`a_Packages`	All loaded packages Sorted based on application All packages are stated in **??**

Tab. 1.2 (continued).

File name (without extension)	Description
b_Meta	Meta information regarding author, title, keywords, etc.
	Boolean variable to select whether it is a dissertation or a bachelor/master thesis
	Boolean variable to select whether it is a master or a bachelor thesis
	Boolean variable to select whether the thesis contains an embargo notice due to a confidentiality agreement; also allows you to set the company's name
	Loads the pdfx package to ensure PDF/A compatibility and to set the metadata of the PDF
c_Commands	Commands regarding format and look of the document
d_Nomenclature	Structure of the List of Symbols
Commands	Optional argument defines the class of a symbol (Latin, Greek, . . .)
	makeindex is used for the generation of the List of Symbols. A short instruction how to run makeindex correctly in Texmaker is given in Sec. 1.8.1.
e_Abbreviation	Definitions of abbreviations in the text
Definitions	Abbreviations can be subdivided into classes
	More information can be found in the acro documentation
f_CodeLanguage	Definition of keyword set and comment commands for a
Specifications	certain programming language
	To apprehend the code with your thesis, either copy the code to an lstlistings environment (see examples in Appendix A) or directly include your files
02_Prematter	
a_Cover	Cover page. Depending on the values of isDiss set in b_Meta.tex, the correct cover is printed. The covers for both thesis and dissertation are shown in Fig. 1.1.
b_Declaration	Declaration that thesis was written honestly
c_Dedication	Dedication for thesis dissertation
	Only included if isDiss is true

Tab. 1.2 (continued).

File name (without extension)	Description
`c_Task`	Scan of the task description Only included if `isDiss` is false
`d_Acknowledgements`	Thanks to important people
`e_Abstract`	Summary in German and English
`f_Publications`	Publications in preparation of the dissertation Only included if `isDiss` is true
`g_Nomenclature`	Definitions of symbols
`h_Abbreviations.tex`	Inclusion of abbreviations
03_Content	
`0_Text`	Loads all chapters
`X_iii`	Single chapter
04_Appendix	
`0_Appendix`	Loads all appendix chapters
`X_Appendix`	Xth appendix
05_Literature_and_Index	
`Bibliography`	Literature database
`myindexstyle`	Style for the Index (see Sec. 1.9.4)

1.6.1 Figures

This folder is added to the `graphicspath` in the `c_Commands.tex` file and contains all figures in this template. Being added to the `graphicspath` means that you may include figures without having to specify a path. However, it might become a little confusing if all your figures are placed within one folder. Luckily, several different folders can be added to the `graphicspath`.

This Is a Very Good Title That Illustrates What This Great Piece of Work Is All About and That Gives an Impression of What a Long Title Looks Like

vorgelegt von
First name second name
aus Berlin

von der Fakultät III – Prozesswissenschaften
der Technischen Universität Berlin
zur Erlangung des akademischen Grades

Doktor der Ingenieurswissenschaften
- Dr.-Ing. -

genehmigte Dissertation

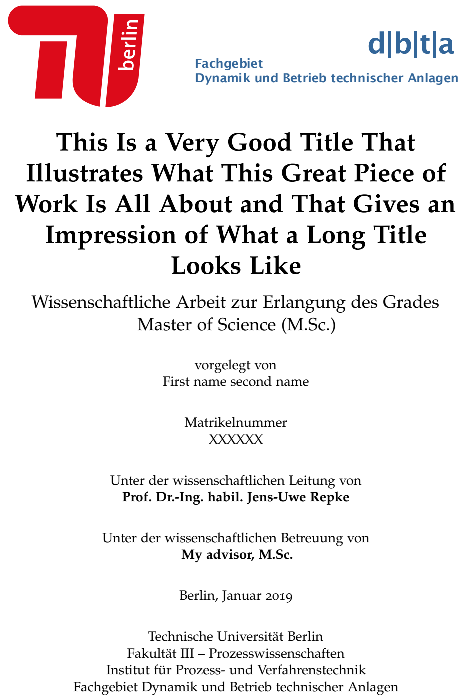
Promotionsausschuss:

Vorsitzende:	Prof. Dr.-Ing. habil. Martina Superwichtig
1. Gutachter:	Prof. Dr.-Ing. habil. Jens-Uwe Repke
2. Gutachter:	Prof. Dr.-Ing. Hans-Georg Wichtig

Tag der wissenschaftlichen Aussprache: 20. Dezember 2018

Berlin, 2019

(a) Cover in diss. mode (`isDiss=true`).



Fachgebiet
Dynamik und Betrieb technischer Anlagen

This Is a Very Good Title That Illustrates What This Great Piece of Work Is All About and That Gives an Impression of What a Long Title Looks Like

Wissenschaftliche Arbeit zur Erlangung des Grades
Master of Science (M.Sc.)

vorgelegt von
First name second name

Matrikelnummer
XXXXXX

Unter der wissenschaftlichen Leitung von
Prof. Dr.-Ing. habil. Jens-Uwe Repke

Unter der wissenschaftlichen Betreuung von
My advisor, M.Sc.

Berlin, Januar 2019

Technische Universität Berlin
Fakultät III – Prozesswissenschaften
Institut für Prozess- und Verfahrenstechnik
Fachgebiet Dynamik und Betrieb technischer Anlagen

(b) Cover in thesis mode (`isDiss=false`).

Fig. 1.1: Covers of this template. These were slightly adapted over the years with newer versions.

1.7 Language: English or German?

The language of this document is set with the `babel` package. The order of the loaded languages determines the default language. Usually, `ngerman` is default (and hence the *second*) language. The `babel` package automatically sets the localized names for Tables and Figures, provides the correct hyphenation, and does more language-related things. In case the thesis is written in English, the order of the languages when loading the `babel` package must be changed. If the language is English, the output decimal marker for SI units (`siunitx` package) is also automatically changed to a period.

1.8 LaTeX Editor Settings and Maintenance

1.8.1 Biber and Makeindex

- command line for setting up `biber` in Texmaker (see Fig. 1.2):

  1) Windows:
     `"C:/path_to/biber.exe" %`
     in TeXLive, `biber` is located in `bin/win32`

  2) Linux:
     `"/usr/path_to/biber" %`

  3) MacOS:
     `"/usr/path_to/biber" %.bcf`

- command line for setting up `makeindex` in Texmaker (see Fig. 1.2):

  1) Windows:
     `"C:/path_to/makeindex.exe" %.nlo -s nomencl.ist -o %.nls`
     in TeXLive, `makeindex` is located in `bin/win32`

  2) Linux:
     `"/usr/path_to/makeindex" %.nlo -s nomencl.ist -o %.nls`

  3) MacOS:
     `"/usr/path_to/makeindex" %.nlo -s nomencl.ist -o %.nls`

Note that `path_to` depends on your system. If the LaTeX distribution is in your system path, simply writing `biber` or `makeindex` (without extension) instead of the whole path should suffice. If you use another editor than Texmaker, check your editor's documentation to find out how to run `biber` and `makeindex` in this software. However, the paths/commands should be similar to those above.

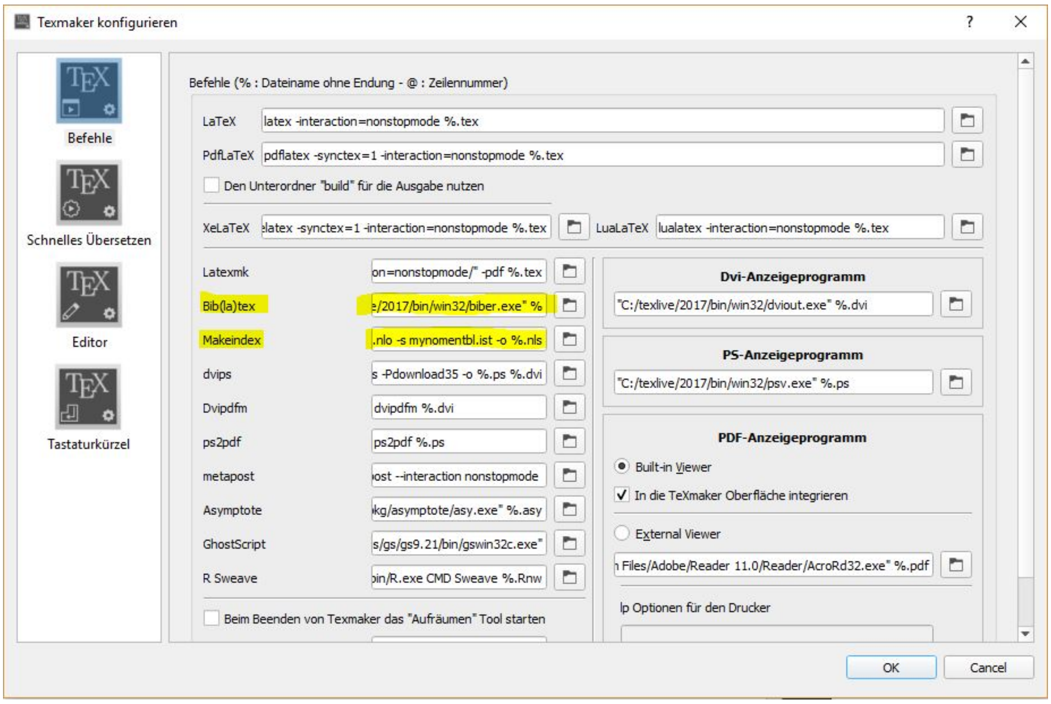


Fig. 1.2: Setting up `biber` and `makeindex` in Texmaker. Zoom in if you need more details.

1.8.2 Compiling the Document

The following commands/scripts must be run to compile the document completely:

1) PDFLaTeX: This generates a first PDF. At this point, the List of References and the List of Symbols are missing.

2) `biber`: Run `biber` (Texmaker standard short key: F11) after you set it up according to Sec. 1.8.1. This creates the necessary temporary reference file.

3) `makeindex`: Run `makeindex` (Texmaker standard short key: F12) after you set it up according to Sec. 1.8.1. This creates the necessary temporary nomenclature file.

4) PDFLaTeX (three times): The first run should already generate the List of References and the List of Symbols. The second run should update all citations etc. in the PDF. Sometimes, a third run is necessary if some references within

the text have changed again. In this case, a third run is necessary. Check the output of Texmaker.

1.8.3 Updating Packages in TEXLive and MiKTEX

As stated above, compilation problems appeared with this template when old versions of the packages were used. Unfortunately, there does not seem to be a possibility to automatically check for updates of packages when they are included in a LATEX document.
In case you have issues when compiling this document, start with updating all your packages. The following instructions are valid for a Windows operating system.

TEXLive manages the packages tlshell. Its GUI is located in `texlive/year/bin/win32`. In this folder, you find a file called `tlshell.exe`. Use this program to update your packages.

MiKTEX manages the packages in the MiKTEX console. Use this application to update your packages.

Similar applications are available on all other operating systems. Restart your LATEX editor after updating all packages and see whether you can compile this template without errors. If your problem persists, please add an issue on the gitlab webpage of this template.[11]

1.9 Extra Features

The following sections introduce a few extra features. None of them are necessary for a thesis (except for PDF/A for a dissertation), but they might be of use for some people.

[11] https://git.tu-berlin.de/dbta/templates/Thesis_template/issues, February 2021

1.9.1 Compatibility with PDF/A

Dissertations are stored as PDF/A at the university library. To ensure compatibility of this template with PDF/A, the `pdfx` package is used with the option `a-2b`. The compiled PDF was successfully validated with Callas[12], as recommended by the university library[13]. Note that contrary to the descriptions given in the linked document in footnote 13, a specific color profile is *not* necessary if you use an up-to-date version of the `pdfx` package as it automatically includes a free color profile. Check the documentation of the `pdfx` package to find how to include other color profiles. Normally, the default should however suffice. The `pdfx` package also loads the `hyperref` package. **Note that the `pdfx` package cannot ensure that all your included PDFs fulfill the PDF/A standard. Hence, check your PDF for PDF/A compatibility whenever you included external files.**

1.9.2 Support for Overleaf or other online editors

This template is compatible with Overleaf[14], an online editor for LaTeX documents. If you would like to write in such an online environment, upload the content of the subfolder `LaTeX_template_thesis` to an empty project and compile it online. This way, you do not have to install any LaTeX distribution locally. **Note that this is not advised if you have sensitive data, for example from a company. In this case, your data should remain on your or your company's computer!**

1.9.3 Automatic Indenting of .tex Files

Indenting *can* be used for structuring one's document, e.g., by indenting everything within an `equation` environment, but is certainly *not* a must-have for a thesis. If you do not need automatic indenting, you can skip this section.
Unfortunately, LaTeX does not offer automatic indenting as do, for example, Matlab

[12] https://conversion.ub.tu-berlin.de/, December 2018
[13] https://www.static.tu.berlin/fileadmin/www/10002444/Dokumente/Forschen_Publizieren/Dissertationsstelle/pdf-guide-202211-de.pdf, October 2025
[14] https://de.overleaf.com/, January 2019

or Python. However, the perl script-based `latexindent`[15] can be used for automatic indenting of the source code. This executable is part of every LaTeX distribution. There are two ways of using `latexindent`: directly running `latexindent` or calling it via `arara`[16]. The arara software is also part of every LaTeX distribution and can be used for TeX automation.[17] The first option is described here for a Windows operating system (the instructions for Linux and Mac should be similar; only the file extension .exe is propably different). With this option, it is only possible to indent one file at a time (the file that is currently open in Texmaker). The second option can automatically indent all files in the document and is described in **??**.
To get automatic indenting, add a user command in Texmaker (User → User Commands → Edit User Commands). You might call one menu item `latexindent`. The command is shown in Code 1.1. This means that `latexindent` is executed with writing rights (`-w`) on the current `.tex` file with local settings (`-l`) in the given path. These local settings are the only issue, because the absolute path can of course change if you move your folder. However, the relative path would always change depending on which file you actually want to indent.
The local settings are important as they specify that only one backup is created. For more information, please refer to the documentation of `latexindent`. You can then execute your new user command in Texmaker with the opened file in question. Afterwards, update your file by clicking on File → Reload document from file. The result is shown in Fig. 1.3. **Note that automatic indenting might not be available for online editors.**

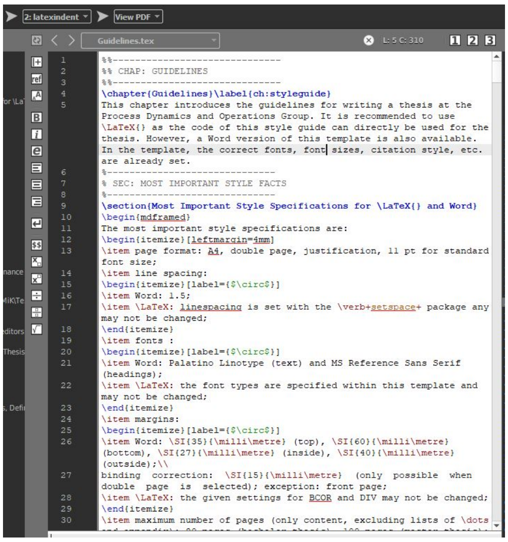
```
"C:/path_to_texlive/year/bin/win32/latexindent.exe" -w %.tex -l="
    absolute_path_to_thesis_template/Thesis_template/
    LaTeX_template_thesis/00_Arara_and_Latexindent/localSettings.yaml"
```

Code 1.1: Setting up `latexindent` in Texmaker.

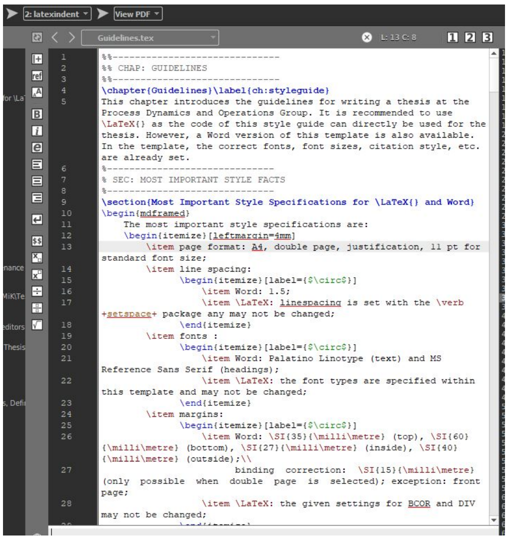
1.9.4 Index

If you want to give the reader the possibility to quickly scan your document for the interesting keywords, you can generate an Index at the end of your thesis. This

[15] https://github.com/cmhughes/latexindent.pl, February 2019
[16] https://tex.stackexchange.com/questions/126241/autoindent-in-texmaker, January 2019
[17] https://github.com/cereda/arara, February 2019



(a) Before.



(b) After.

Fig. 1.3: Before (left) and after (right) automatic indenting with `latexindent`.

is certainly not necessary for a bachelor or master thesis, but can be used for a dissertation. In this template, the `imakeidx` package is used. This automatically executes `makeindex` during compilation. If you do not want an Index, remove the `\makeindex` command from the `c_Commands.tex` file and the `\printindex` command from the `main.tex` file.

The Index works as follows: You simply write your text and add the keyword to the index with the `\index{}` command. You can combine keywords to categories, e.g., Thermodynamics might be one keyword. Now you can add other keywords to this category with `\index{keyword!subkeyword}`. This could be equations of state or activity models. The results are shown in the Index on page 61.

1.9.5 Software for Vector Graphics

Figures are an important part of every thesis. Normally, these figures are generated by the author, who is hence responsible for their quality. Vector graphics are generally preferred to raster graphics (JPG, PNG, ...)

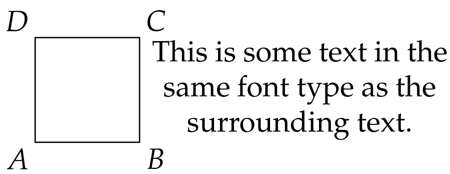


Fig. 1.4: Image generated with Asymptote.

as their quality does not depend on the resolution. There are a few programs for generating figures within LaTeX, which are listed in the following. Plotting within LaTeX has a few advantages, for example: the figure always has the same font type as the text, and global settings can be applied to all figures within one document. Typically, one can either compile all figures whenever PDFLaTeX is executed, or only compile the changed figures, or generate the PDFs in a separate file and include the PDF in the document.

Asymptote is a powerful vector graphics program[18]. It can be made visible to all LaTeX distributions with a package[19]. Asymptote can directly generate a PDF output. An example of an image generated with Asymptote is shown in Fig. 1.4. In this case, the image was generated externally and is now included as a PDF. The code of this image is located in the `Examples` folder of this template.

TikZ is another vector graphics tool[20]. It is also available for all LaTeX distributions.[21] Asymptote and TikZ are similar as all images are produced based on commands (similar to LaTeX code). Choose between these two programs based on your own preferences.

In TikZ, it is recommended to externalize your figure, i.e., they are only redrawn if something was changed. In this case, add the lines in Code 1.2 *below* the section on makeindex in the `c_Commands.tex` file to make sure everything works smoothly. First of all, the external library is loaded, secondly, externalization is started. Importantly, shell escape must be enabled here. In addition, the `\includepdf` command is excluded from externalization. Finally, the `\todo` command is redefined so it also disables externalization locally. More on these issues can be found here[22,23]. **You need to create the folder `TikZ` if you use this prefix**.

[18] http://asymptote.sourceforge.net/, March 2019
[19] https://ctan.org/pkg/asymptote, March 2019
[20] https://sourceforge.net/projects/pgf/, March 2019
[21] https://www.ctan.org/pkg/pgf, March 2019
[22] https://tex.stackexchange.com/questions/135504/includepdf-causes-an-error-message-from-pgfplots-externalization, August 2019
[23] https://tex.stackexchange.com/questions/42486/todonotes-and-tikzexternalize, August 2019

PSTricks was designed for PostScript vector graphics[24] and is available for all LaTeX distributions.[25] Due to its PostScript origin, it cannot directly be used with PDFLaTeX. If you want to compile with PDFLaTeX, the best solution is using the package `auto-pst-pdf`[26].

Inkscape is a graphics tool, which is not used with commands (as are the others), but which is similar to Visio or Powerpoint in its use.[27] However, Inkscape exports an additional file containing the LaTeX specifiations for an image. Hence, the text of an exported image will adapt to the font size and type of the LaTeX document.

Draw.io is an open source program for drawing flowcharts or pipe & instrumentation diagrams.[28] It contains many more shapes for process engineering than a standard version of Microsoft Visio. Figures can be exported as PDF and further, for example, further processed in Inkscape. Draw.io is available as browser application or can be installed on a computer.

mathcha.io is an online LaTeX-based editor and may be used for generating small sketches and drawing.[29] Designed graphs can be exported to TikZ or as images.

```
\usetikzlibrary{external}
\tikzexternalize[prefix=Tikz/,shell escape=-enable-write18,optimize
    command away=\includepdf]
\tikzset{external/system call={pdflatex \tikzexternalcheckshellescape -
    halt-on-error -interaction=batchmode -jobname "\image" "\texsource"}}
\makeatletter
\renewcommand{\todo}[2][]{\tikzexternaldisable\@todo[#1]{#2}\
    tikzexternalenable}
\makeatother
```

Code 1.2: Command lines for externalization in TikZ.

[24] http://www.tug.org/PSTricks/main.cgi/, March 2019
[25] https://www.ctan.org/pkg/pstricks-base, March 2019
[26] https://ctan.org/pkg/auto-pst-pdf?lang=de, March 2018
[27] https://inkscape.org/de/, March 2019
[28] www.draw.io, June 2020
[29] https://www.mathcha.io, August 2020

1.10 Templates for Typical Elements of a Thesis

This section introduces templates for the typical elements of a thesis, such as tables, figures, equations, etc.

1.10.1 Tables

A template for a Table is given in Tab. 1.4. Table 1.4 should not be abbreviated at the beginning of a sentence. A common issue in LaTeX are footnotes within tables. However, there is the `threeparttable` package to deal with this. In such a `threeparttable` environment, annotations can be easily added (Tab. 1.5). Normal tables in LaTeX cannot go over several pages. For longer tables, see the `longtable` package in **??**. This template also loads the `threeparttablex` package, which extends the annotation feature to the `longtable` environment. The `wrapfig` package can also be used for wrapped tables as shown in Tab. 1.3. Again, make sure that the Table is small enough.

Tab. 1.3: A wrapped table.

Variable	Mean	Std. Dev.
a	4	± 0.1

Tab. 1.4: This is the caption of the Table in the text. Is is placed *above* the table. It can be longer and contain additional information. Vertical lines should be avoided in tables. A full stop is automatically added after the last sign.

Entry 1 Unit 1	Entry 2 Unit 2	Entry 3 Unit 3	Unit column Some text	*Italics* *Some text*
1	2	3	J	*bla*
4	5	6	$\mathrm{Pa\,m^{-2}}$	*bla bla*

Tab. 1.5: Exemplary table with an annotation.

Variable	Mean	Std. Dev.
a	4	$\pm 0.1^{*}$

* This standard deviation is only true if I measured correctly

1.10.2 Figures

A template for a Figure is given in Fig. 1.5. Figure 1.5 should not be abbreviated at the beginning of a sentence. Note that the often used `\ref` command is not used here. Instead, Sec. 1.10.10 introduces the `\autoref` command.

d|b|t|a
Fachgebiet
Dynamik und Betrieb technischer Anlagen

Fig. 1.5: This is the caption of the Figure in the text. Is is placed *below* the Figure. It can be longer here and contain additional information, such as references or keys for the graphs. Note that one-line captions are justified. A full stop is automatically added after the last sign.

berlin

Fig. 1.6: A wrapped figure.

Although it is not seen on a regular basis in theses or dissertations, a Figure may also be surrounded by text. The `wrapfig` package is used for this purpose. It depends on the Figure's size if this is a good or a bad idea. Make sure that readability of the Figure is still given. An example is given in Fig. 1.6. Another possibility is putting descriptive text in an otherwise raw figure. This is illustrated in Fig. 1.7. The `overpic` package provides the necessary environment of the same title and the `\put` command to add arbitrary text. If the environment is loaded with the additional options `tics=10, grid`, a grid with ten tics is drawn to ease the positioning of the text. Thus, the inserted text automatically uses the same font type as the surrounding text.

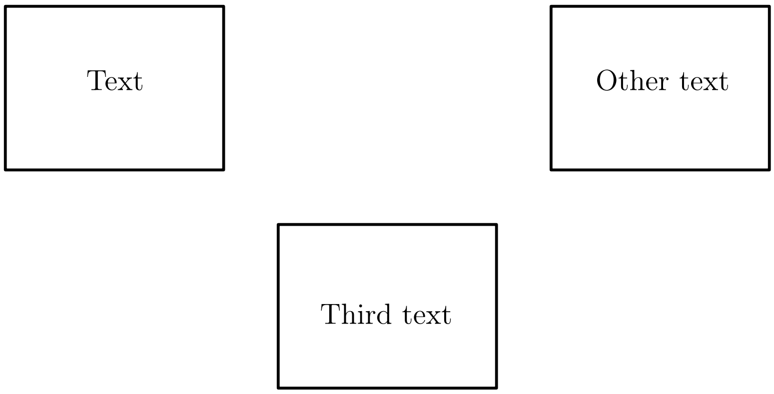


Fig. 1.7: Example of overpic environment.

1.10.3 Numbers and Units

Numbers and units are very important, and there are some rules when typesetting them. For example, units are never written in italics. They should also have the right space between them. For this purpose, the `siunitx` package is suggested.

- numbers (`\num{number}`): 3.141 59
- exponentials (`\num{numbere6}`): 1.3×10^{6}
- units (`\unit{\unit}`): $\mathrm{J\,mol^{-1}\,K^{-1}}$
- numbers+units (`\qty{number}{\unit}`): $8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$
- ranges (`\qtyrange{number1}{number2}{\unit}`): 4 to 10 K.
- uncertainty (`\num{number(uncertainty)}`): 410.33 ± 0.55
- uncertainty with units (`\unit{number(uncertainty)}{\unit}`): $(410.33 \pm 0.55)\,\mathrm{J}$
- own units can also be defined as has been done for $\mathrm{kJ\,mol^{-1}}$ and $\mathrm{kJ\,mol^{-1}\,K^{-1}}$

1.10.4 Equations

An exemplary Equation is given in Eq. (1.1). Equation (1.1) should not be abbreviated at the beginning of a sentence. The efficient used commands to generate the partial derivative were made with the `xparse` package.

$$\left(\frac{\partial^2 f}{\partial x^2}\right) = \left[\frac{\partial^4 g}{\partial x^4}\right]. \tag{1.1}$$

For important equations, you might want to use a box:

$$\boxed{E = mc^2} \tag{1.2}$$

1.10.5 Optimization problems

In many applications, optimization problems must be stated in theses and dissertations. This can be realized with the `optidef` package:

$$\begin{aligned} \min_{\omega} \quad & f(\omega) \\ \text{s.t.} \quad & 0 \geq g(\omega) \quad \text{(Constraints)}, \\ & \omega \in \Omega\,. \end{aligned} \tag{1.3}$$

1.10.6 Acronyms and Abbreviations

Abbreviations are defined in `e_AbbreviationDefinitions.tex` using the `acro` package. New abbreviations must typically be explained at their first appearance in the text. The `\ac` command uses the defined acronyms (see List of Abbreviations) for doing that. For example, the ***Su****ite of* ***n****onlinear and* ***di****fferential-***al****gebraic equation* ***s****olvers* (SUNDIALS) is explained here, but not afterwards because SUNDIALS was already defined. It seems to be more to write, but thus you make sure that an abbreviations is explained only at its first appearance ... even if you change your text completely. In addition, you link your abbreviations to the List of Abbreviations. The `acro` package also contains specific commands for the plural of the long and the short form of the abbreviations; check the documentation for

more information. Moreover, the abbreviations are automatically added to the Index. Note that expressions, such as e.g., or i.e., *should not* be added to the List of Abbreviations.
You can define several tags with the acro package as has here been done for ***D****ifferential-****a****lgebraic* ***e****quation (system)* (DAE) or ***a****ctive* ***b****ound* (ab).

1.10.7 References

LaTeX – or more precisely biber – includes literature if it is stored as .bib file. It is however not recommended to manually write in .bib files, but one should use a program for administrating literature. Examples of such programs are Mendeley or Jabref (Fig. 1.8). In Jabref, one can add literature via the DOI or the ISBN. Furthermore, it offers templates for all standard document classes, such as articles, books, online references, and more.
In the following, a few examples of the authoryear short reference are stated. For more information, the reader is referred to the documentation of the biblatex package. Biblatex and biber are used because they are compatible with UTF8. Hence, Umlaute, such as ä, do not have to be rewritten as was the case in bibtex. In addition, biblatex supports editing of the citation style via TeX and LaTeX commands. Hence, the tedious editing of bibliography styles (.bst files), which were used with bibtex, is not necessary anymore.

Note that references to books should contain the page number.

- book: (**Coker2007**), **Coker2007**
- article: (**Abrams1975**), **Abrams1975**
- conference paper: (**Penteado2018**), **Penteado2018**
- online: (**benzene_nist2017**), **benzene_nist2017**
- dissertation: (**Cuda2012**), **Cuda2012**
- bachelor/master thesis: (**Hoffmann2015**), **Hoffmann2015**

These commands are used as

- \textcite{bibtexkey}:
  "**Abrams1975** stated that thermodynamics are great."

- \parencite{bibtexkey}:
  "Thermodynamics are great (**Abrams1975**)."

Another helpful method are shorthands. These can be used when the original author or institution would lead to a very long reference in the text, e.g., for norms. In the following, the normal reference and the shorthand citation are shown. The \textcite{} has been redefined to introduce the shorthand:

- **DIN1319_4**

- **DIN1319_4**

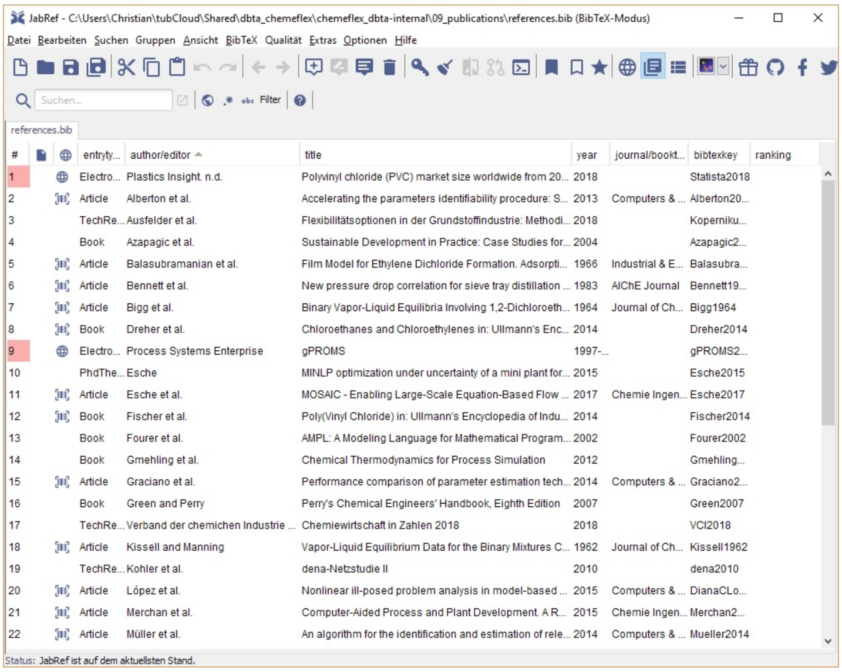


Fig. 1.8: Jabref.

1.10.8 Chemistry and Chemical Reactions

Sometimes, it is necessary to state chemical reactions or molecules. For this purpose, the chemfig and the chemformula package, which is loaded as part of the

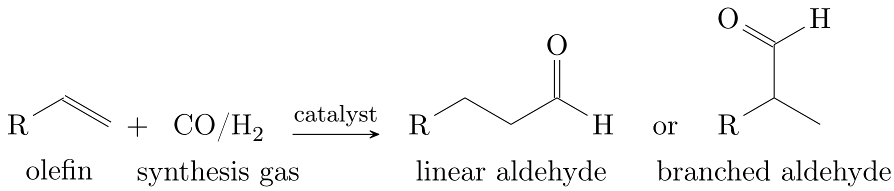


Fig. 1.9: Hydroformylation reaction scheme to demonstrate the two chemistry packages.

chemmacros package, are used as shown in Fig. 1.9. Molecular formulas should not be written in math mode, but can be typeset with the \ch{} command, e.g., H_2O. Note that chemical formulae, e.g., CO_2 for carbon dioxide, *should not* be added to the List of Symbols or the List of Abbreviations, while abbreviations, e.g., MEA for monoethanolamine, *should* be added to the List of Abbreviations. Greek letters in chemicals are not typeset in italics, therefore the upgreek package is loaded to typeset γ-Aluminium instead of γ-Aluminium.

1.10.9 Theorems, Lemmas, Proofs, Remarks, Definitions, and Algorithms

All of these items are introduced in the following. Note that frames and their colors are arbitrary. You might want to change the color or the linewidth. You can do this in the c_Commands.tex file.

Theorems: Theorems can be defined using the amsthm package in combination with the mdframed package for a possible frame. An exemplary Theorem is given in Theorem 1.1.

Theorem 1.1 (What is theoremed): *Let f be a function whose derivative exists in every point, then f is a continuous function.*

Lemmas: Lemmas can be defined using the amsthm package in combination with the mdframed package for a possible frame. An exemplary Lemma is given in Theorem 1.2.

Lemma 1.2 (What needs to be lemma'd): *Given two line segments whose lengths are a and b, respectively, there is a real number r such that $b = ra$.*

Proofs: Proofs can be defined using the `amsthm` package. They are typically not numerated as they follow a certain Theorem or Lemma.

What needs to be proven. To prove it by contradiction, try and assume that the statement is false, proceed from there, and at some point, you will arrive at a contradiction. □

Remarks: Remarks can be defined using the `amsthm` package in combination with the `mdframed` package for a possible frame. An exemplary Remark is given in Theorem 1.3.

Remark 1.3 (What the remarker remarks): *This statement is true, I guess.*

Definitions: Definitions can be defined using the `amsthm` package in combination with the `mdframed` package for a possible frame. An exemplary Definition is given in Theorem 1.4. Equation (1.4) can also be referenced.

Definition 1.4 (What the definition defines): *This is a definition. It defines itself.*

$$c = \infty \tag{1.4}$$

Algorithms: Algorithms can be displayed using the `algorithm2e` package. An example is shown in Alg. 1.1.

1.10.10 Autorefs

The `hyperref` package also supplies an `\autoref` command, which is linked to babel. In this case, Fig. or Tab. are localized and you do not have to remember whether you used Fig. or Figure or something else in the text. As stated above, the long version of an item should be used at the beginning of a sentence (the `\Autoref{label}` command was defined for this purpose). Some short forms are equal to their long forms as there is not really a good abbreviation:

Algorithm 1.1: How to write algorithms

```
Data: this text
Result: how to write algorithm
initialization;
while not at end of this document do
    read current;
    if understand then
        go to next section;
        current section becomes this one;
    else
        go back to the beginning of current section;
    end
end
```

1) German ...
   - ..., siehe Kap. 1. Kapitel 1 zeigt, dass ...
   - ..., siehe Abschn. 1.10. Abschnitt 1.10 zeigt, dass ...
   - ..., siehe Abschn. 1.8.1. Abschnitt 1.8.1 zeigt, dass ...
   - ..., siehe Abb. 1.5. Abbildung 1.5 zeigt, dass ...
   - ..., siehe Tab. 1.4. Tabelle 1.4 zeigt, dass ...
   - ..., siehe Gl. (1.1). Gleichung (1.1) zeigt, dass ...
   - ..., siehe Satz 1.1. Satz 1.1 zeigt, dass ...
   - ..., siehe Satz 1.2. Satz 1.2 zeigt, dass ...
   - ..., siehe Satz 1.3. Satz 1.3 zeigt, dass ...
   - ..., siehe Satz 1.4. Satz 1.4 zeigt, dass ...
   - ..., siehe Alg. 1.1. Algorithmus 1.1 zeigt, dass ...
2) English ...
   - ..., see Chap. 1. Chapter 1 shows that ...

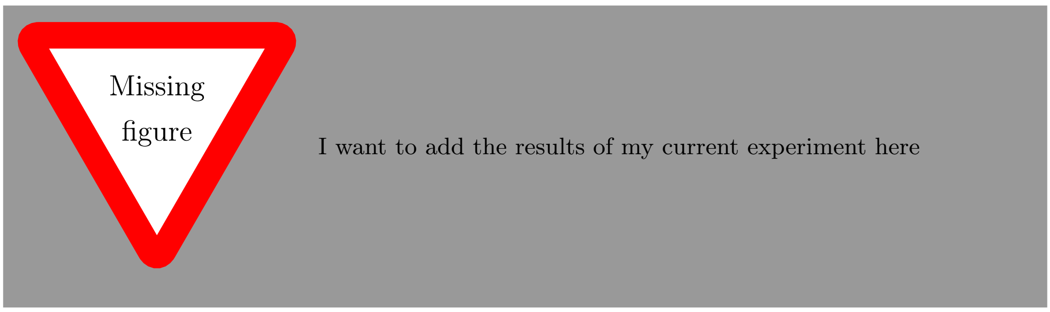
- . . . , see Sec. 1.10. Section 1.10 shows that . . .
- . . . , see Sec. 1.8.1. Section 1.8.1 shows that . . .
- . . . , see Fig. 1.5. Figure 1.5 shows that . . .
- . . . , see Tab. 1.4. Table 1.4 shows that . . .
- . . . , see Eq. (1.1). Equation (1.1) shows that . . .
- . . . , see Theorem 1.1. Theorem 1.1 shows that . . .
- . . . , see Theorem 1.2. Theorem 1.2 shows that . . .
- . . . , see Theorem 1.3. Theorem 1.3 shows that . . .
- . . . , see Theorem 1.4. Theorem 1.4 shows that . . .
- . . . , see Alg. 1.1. Algorithm 1.1 shows that . . .

1.10.11 ToDos

change that asap!

change that later at some point

In your thesis, you always have parts where you will have to do something more later on. You can mark these sections with the `\todo[option]{text}` command. Something that should be done later might be green. You can also include a "missing figure" if you currently do not have it. When you do not have any more ToDos, remove the List of ToDos from the `main.tex`.



2 Introduction

Some introducing words ...

2.1 Motivation

Give a general overview on the subject of this thesis! In which context shall the content of this thesis be seen?

2.2 Research Goal

What questions shall be investigated and answered in this thesis? What is the scope of this work?

2.3 Outline of Work

What is the structure of this thesis?

3 Étalé Morphisms

Étalé morphisms are the central class of morphisms in this thesis, since they form the basic building blocks of the étale Grothendieck topology on schemes. Before we dive deeper into general schemes, we first recall the relevant notions for rings and affine schemes.

Definition 3.1 (Thickening): *(Görtz, Wedhorn, 2023, (Definition 18.1), p. 32) Let $\iota : Y \hookrightarrow X$ be a closed immersion of a quasi-coherent ideal $\mathscr{I} \subset \mathscr{O}_X$. We call ι a thickening, iff $\mathscr{I}$ is locally nilpotent, i.e. there exists en open cover $(U_i)_{i\in I}$ of X and $n_i \in \mathbb{N}$, such that $(\mathscr{I}|_{U_i})^{n_i+1} = 0$. For $n \in \mathbb{N}$ we call ι a thickening of order n, if $\mathscr{I}^{n+1} = 0$. A thickening of order 1 is called a first order thickening.*

Definition 3.2 (Étalé ring maps): *(Görtz, Wedhorn, 2023, (18.4), p. 36) We call an R-algebra A formally étalé, iff for all rings C and all ideals $I \lhd C$ the following diagram admits a unique morphism $A \to C$.*

$$\begin{tikzcd} C/I & A \arrow[l] \arrow[dl, dashed] \\ C \arrow[u] & R \arrow[l] \arrow[u] \end{tikzcd}$$

Definition 3.3 (Formally étalé morphism of affine schemes): *Analogously to definition 3.2 we define a morphism of affine schemes* $\operatorname{Spec}(A) \to \operatorname{Spec}(B)$ *to be formally étalé, iff for all closed immersions* $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$

it admits a unique lift fitting into this commutative diagram:

$$\begin{tikzcd} \mathrm{Spec}(R/I) \arrow[r] \arrow[d] & \mathrm{Spec}(A) \arrow[d] \\ \mathrm{Spec}(R) \arrow[r] \arrow[ur, dashed] & \mathrm{Spec}(B) \end{tikzcd}$$

This naturally leads to the definition of formally étalé morphisms of schemes:

Definition 3.4 (Formally étalé morphism of schemes): *(Görtz, Wedhorn, 2023, (Definition 18.3), p. 32) A morphism of schemes $X \to S$ is called formally étalé, iff for all first order thickenings $T_0 \to T$ with affine T, there exists a unique lift fitting into this diagram:*

$$\begin{tikzcd} T_0 \arrow[r] \arrow[d] & X \arrow[d] \\ T \arrow[r] \arrow[ur, dashed] & S \end{tikzcd}$$

Proposition 3.5 : *Formally étalé morphisms are stable under composition and base change.*

Theorem 1 (Locality of étalé morphism): *(Görtz, Wedhorn, 2023, Theorem 18.42, p. 44) Let $f : X \to S$ be a morphism of schemes locallly of finite presentation. Let $x \in X$ and $s := f(x)$. Then f is étalé at x, iff there exists an open affine $V = \mathrm{Spec}(R)$ of s and an open affine $U = \mathrm{Spec}(A) \subset f^{-1}(V)$ of x, such that A is isomorphic to the standard étalé R-algebra.*

Proof. Inhalt... □

4 Grothendieck Topology

In this chapter we want to define a Grothendieck topology on a category. Given a topological space X, the open sets define a category, where the objects are the open sets of X and morphisms are given by inclusion. In algebraic geometry one tries to analyze sheaves of the space X. The category associated to X encodes the coverings of open subest of X by opens.
The notion of a Grothendieck topology will generalize this for arbitrary categories and arbitrary "coverings"

Definition 4.1 (Sieve): *(Mac Lane, Moerdijk, 1992, p. 38) Let $X \in \mathcal{C}$ be an object of a small category. A sieve on X is a subfunctor of the representable presheaf $\mathrm{Hom}_{\mathcal{C}}(-, X)$. We will write h_X for the Yoneda embedding $\mathrm{Hom}_{\mathcal{C}}(-, X)$.*

Definition 4.2 (Grothendieck Topology): *(Mac Lane, Moerdijk, 1992, (Definition 1), p. 110) A Grothendieck topology J on a small categrory $\mathcal{C}$ is a function that assigns to each object $X \in C$ a collection of sieves $J(X)$ called covering sieves, such that the following axioms are satisfied:*

1) *For each object $X \in \mathcal{C}$ $\mathrm{Hom}_{\mathcal{C}}(-, X)$ is a sieve.*

2) *For each morphism $f : X \to Y$ in $\mathcal{C}$ and each sieve $S \in J(Y)$ the pullback $f^*(S) \in J(X)$ is a sieve on X.*

3) *For each covering sieve $S \in J(Y)$ and each sieve R on Y, such that $f^*(R) \in J(X)$ for all $f : X \to Y \in S$, then $R \in J(Y)$*

Remark 1 : *For a sieve S on $Y \in \mathcal{C}$ and any $f : X \to Y$ in $\mathcal{C}$ we have $f^*S = \{g : Z \to X \mid f \circ g \in S\} \subset \mathrm{Hom}_{\mathcal{C}}(-, X)$*

Definition 4.3 (Site): *(Mac Lane, Moerdijk, 1992, p. 110) For a small category $\mathcal{C}$ and a Grothendieck topology J we will call the pair $(\mathcal{C}, J)$ a site on $\mathcal{C}$.*

Definition 4.4 (Matching family): *(Mac Lane, Moerdijk, 1992, p. 121) Let $(\mathcal{C}, J)$ be a site, $P : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ a presheaf on $\mathcal{C}$ and $S \in J(C)$ a covering sieve. A matching familiy for S of elements of P is natural transformation $\eta : S \Rightarrow P$.*

This assigns to each $f : D \to C \in S(D)$ an element $x_f \in P(D)$, such that $P(g)(x_f) = x_{f \circ g}$ for all $g : E \to D \in \mathcal{C}$. Notice that $f \circ g \in S(E)$, since S is a sieve.

An amalgation of the matching family η is an element $x \in P(C)$, such that $P(f)(x) = x_f$ for all $f \in S$

Definition 4.5 (Sheaf): *(Mac Lane, Moerdijk, 1992, (p. 121) Let $(\mathcal{C}, J)$ be a site. A presheaf $P : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ is a sheaf, iff for every covering sieve $S \in J(C)$ the following diagramm admits a unique lift:*

$$\begin{tikzcd} S \arrow[r] \arrow[d, hook] & P \\ h_C \arrow[ur, dashed] & \end{tikzcd}$$

Write $\mathsf{Sh}(\mathcal{C}, J) \subset \mathsf{PSh}(\mathcal{C})$ for the full subcategory of sheaves.

Remark 2 : *The sheaf condition is a local property.*

Lemma 1 (Sheafication): *(Mac Lane, Moerdijk, 1992, p. 134) Let $(\mathcal{C}, J)$ be a site, then the fully faithful inclusion $\iota : \mathsf{Sh}(\mathcal{C}, J) \hookrightarrow \mathsf{PSh}(\mathcal{C})$ admits a left adjoint $a \dashv \iota$.*

Corollary 1 : *By Mac Lane, 1998, the category* $\mathsf{Sh}(\mathcal{C},J)$ *is complete. In particular the limit of sheaves is again a sheaf and can be calculated in the category* $\mathsf{PSh}(\mathcal{C})$ *of presheaves.*

5 Étalé Site

Definition 5.1 (Étalé covering): *(Görtz, Wedhorn, 2023, (Definition 20.21), p. 101) An étalé covering of a scheme X is a family of étalé morphisms $(g_i : U_i \to X)_{i \in I}$, such that $X = \bigcup_{i \in I} g_i(U_i)$. Let $\mathcal{U} = (U_i \to X)_{i \in I}$ and $\mathcal{V} = (V_j \to X)_{j \in J}$ be étalé coverings, than we call $\mathcal{V}$ a refinement of $\mathcal{U}$, if there exists a map $\alpha : J \to I$, such that for each $j \in J$ there exists a morphism $V_j \to U_{\alpha(j)}$, such that the following commute:*

$$\begin{tikzcd} V_j \arrow[rr] \arrow[dr] & & U_{\alpha(j)} \arrow[dl] \\ & X & \end{tikzcd}$$

Definition 5.2 (Étalé Site): *We write Sch_{et} for the étalé site (Sch, J), where $J(X)$ contains all étalé coverings of X.*

Proposition 5.3 : *(Görtz, Wedhorn, 2023, p. 101) Let X be a scheme. Then every étalé covering $\mathcal{U} = (g_i : U_i \to X)_{i \in I}$ of X admits a refinement $\mathcal{V} = (V_j \to X)_{j \in J}$ of affine schemes. If X quasi-compact, the refinement can be chosen to be finite.*

Proof. For each $(U_i)_{i \in I}$ chose an open affine cover $U_i = \bigcup_{j \in J_i} V_{ij}$. Now consider the composite $V_{ij} \hookrightarrow U_i \xrightarrow{g_i} X$. Each $V_{ij} \hookrightarrow U_i$ is an open immersion, hence étalé. Let $J = \coprod_{i \in I} J_i = \{(i,j) \mid i \in I, j \in J_i\}$. This yields an étalé cover $\mathcal{V} = (V_{ij} \to X)_{(i,j) \in J}$, since the composite of two étalé morohisms is étalé and the image covers X by construction. $\mathcal{V}$ is a refinement of $\mathcal{U}$ via $\alpha : J \to I, (i,j) \mapsto i$. If

X is quasi-compact, there exists a finite subset $J' \subseteq J$ such that the opens $g_{\alpha(j)}(V_j)$ cover X. $\square$

Lemma 2 : *(Atiyah, Macdonald, 1969, Ch. 3, Ex. 16) Let $f : R \to S$ be faithfully flat ring, then f is injective.*

Lemma 3 (Representable presheaves of affine schemes): *Let A be an commutative ring and let $\{\alpha_i : R \to R_i\}_{i \in I}$ be a morphism of étalé ring maps, such that the induced morphism of schemes*

$$\coprod_{i \in I} \mathrm{Spec}(R_i) \to \mathrm{Spec}(R)$$

is surjective, i.e. an étalé covering.
Then the sequence

$$\mathrm{Hom}(A, R) \longrightarrow \prod_i \mathrm{Hom}(A, R_i) \rightrightarrows \prod_{i,j} \mathrm{Hom}(A, R_i \otimes_R R_j)$$

is exact, where the two parallel morphisms are induced by $R_i \to R_i \otimes_R R_j$,$r \mapsto r \otimes_R 1$ and $R_j \to R_i \otimes R_j, r \mapsto 1 \otimes_R r$.

Proof. Since $\mathrm{Spec}(R)$ is quasi compact, by 5.3 we can refine the cover to a finite cover. So without loss of generality we assume I to be finite.

Assume we are given $\varphi_i : A \to R_i$, such that

$$\varphi_i \otimes_R 1 = 1 \otimes_R \varphi_j : A \to R_i \otimes_R R_j \tag{5.1}$$

for all i,j, i.e. the following commutes.

$$\begin{tikzcd} & R_i \arrow[dr, "id \otimes_R 1"] & \\ A \arrow[ur, "\varphi_i"] \arrow[dr, "\varphi_j"'] & & R_i \otimes_R R_j \\ & R_j \arrow[ur, "1 \otimes id"'] & \end{tikzcd}$$

We will construct a morphism unique $\varphi : A \to R$, such that $\alpha_i \circ \varphi = \varphi_i$, i.e. the following commutes:

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[dr, "\varphi_i"'] & R \arrow[d, "\alpha_i"] \\ & R_i \end{tikzcd} \tag{5.2}$$

The $\varphi_i : A \to R_i$ induce a morphism $\Phi : A \to S$ given by $\Phi(a) = (\varphi_i(a))_i$. Let $S := \prod_i R_i$ and $\alpha : R \to S$ be the canonical ring map induced by the α_i, i.e. $\alpha(r) = (\alpha_i(r))_i$. By The Stacks Project Authors, 2024, Tag 00HQ, a flat ring map $R \to S$ is faithfully flat if and only if $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is surjective. Since the R_i cover R, and S is free, hence flat, the induced map $R \to S$ is faithfully flat.

Let $\delta_1, \delta_2 : S \to S \otimes_R S$ be given by $\delta_1 = id \otimes_R 1, \delta_2 = 1 \otimes_R id$. Since, the cover is finite, S is finitely generated. Hence we obtain a natural isomorphism

$$S \otimes_R S \cong \big(\prod_i R_i\big) \otimes_R \big(\prod_j R_j\big) \cong \prod_i \big(R_i \otimes_R \prod_j R_j\big) \cong \prod_{i,j} \big(R_i \otimes_R R_j\big)$$

The morphisms δ_1, δ_2 induce $\tilde{\delta_1}, \tilde{\delta_2} : S \to \prod_{i,j} R_i \otimes_R R_j$ defined by $\tilde{\delta_1}((r_i)_i) = (r_i \otimes_R 1)_{i,j}$ and $\tilde{\delta_2} = (1 \otimes_R r_j)_{i,j}$. Now precomposing with Φ yields

$$\tilde{\delta_1}(\Phi(a)) = (\varphi_i(a) \otimes_R 1)_{i,j}$$
$$\tilde{\delta_2}(\Phi(a)) = (1 \otimes_R \varphi_j(a))_{i,j}$$

Now the compatibility assumption 5.1 yields

$$\delta_1 \circ \Phi = \delta_2 \circ \Phi \tag{5.3}$$

Consider the following diagram:

$$0 \longrightarrow R \xrightarrow{\alpha} S \xrightarrow{\delta_1 - \delta_2} S \otimes_R S \tag{5.4}$$

Claim: This is exact, i.e. $R \cong ker(\delta_1 - \delta_2) \cong Eq(\delta_1, \delta_2)$.

By lemma 2 α is injective. Let $r \in R$, then $\delta_1(\alpha(r)) = \alpha(r) \otimes_R 1 = 1 \otimes_R \alpha(r) = \delta_2(\alpha(r))$, so $im(\alpha) \subset Eq(\delta_1, \delta_2)$.

Now assume $s \in ker(\delta_1 - \delta_2)$, i.e. $s \otimes_R 1 = 1 \otimes_R s$. Let $\bar{s}$ be the image in $\pi : S \to S/\alpha(R)$. Tensoring π with S yields $\pi \otimes_R id : S \otimes_R S \to (S/\alpha(R)) \otimes_R S$. Next apply this to $s \otimes_R 1 = 1 \otimes_R s$ yields

$$\bar{s} \otimes 1 = (\pi \otimes_R id)(s \otimes_R 1) = (\pi \otimes_R id)(1 \otimes_R s) = \pi(1) \otimes_R s = 0 \otimes s = 0.$$

So $\bar{s} \otimes_R 1 = 0$ in $(S/\alpha(R)) \otimes_R S$. Now since α is faithfully flat,

$$S/\alpha(R) \to (S/\alpha(R)) \otimes_R S$$

is injective. Hence $\bar{s} = 0$, i.e. $s \in \alpha(R)$, which proves the claim.

Now by 5.3 and by exactness of 5.4, the universal property of the kernal R induces a unique morphism $\varphi : A \to R$, such that the following commutes:

$$\begin{tikzcd} & & A \arrow[dl, dashed, "\varphi"'] \arrow[d, "\Phi"] \arrow[dr] & \\ 0 \arrow[r] & R \arrow[r, "\alpha"] & S \arrow[r, "\delta_1 - \delta_2"'] & S \otimes_R S \end{tikzcd}$$

Hence, $(\alpha_i(\varphi(a))_i = \alpha(\varphi(a)) = \Phi(a) = (\varphi_i(a))_i$, which proves commutativity of 5.2.

$\square$

This allows us to restrict to finite affine étalé covers. Such reductions are standard in algebraic geometry: many statements can then be checked on affines and translated into commutative algebra.

Theorem 2 (Representable presheaves of schemes): *Let X be a scheme. Consider the representable presheaf*

$$h_X : (\mathsf{Sch}_{/Et})^{\mathrm{op}} \to \mathsf{Set}$$

on the étalé site of Sch. *Then h_X is a sheaf in the étalé topology.*

Proof. We first assume X to be an affine scheme. Let U be a scheme and $\mathcal{U} = (U_i \to U)_{i \in I}$ be an étalé covering. By 5.3 without loss of generality we may assume the U_i to be affine. We need to show, that the following diagram is exact:

$$h_X(U) \longrightarrow \prod_i h_X(U_i) \rightrightarrows \prod_{i,j} h_X(U_i \times_U U_j)$$

Obviously the composite is equal by the pullback property.

We will first reduce U to the affine case. Let $U = \bigcup_\alpha V_\alpha$ be an affine open cover. Assume $f_i : U_i \to X$, such that $U_i \times_U U_j \to U_i \xrightarrow{f_i} X$ and $U_i \times_U U_j \to U_j \xrightarrow{f_j} X$ coincide, i.e. the following commutes:

$$\begin{tikzcd} U_i \times_U U_j \arrow[r] \arrow[d] & U_j \arrow[d, "f_j"] \\ U_i \arrow[r, "f_i"'] & X \end{tikzcd} \tag{5.5}$$

To construct a unique morphism $f : U \to X$ it suffices to construct compatible morphisms $g_\alpha : V_\alpha \to X$ for all α and then glue Zariski locally.

Consider the following pullback diagram.

$$\begin{tikzcd} V_\alpha \times_U U_i \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & U_i \arrow[d] \\ V_\alpha \arrow[r] & U \end{tikzcd}$$

The morphisms $V_\alpha \times_U U_i \to V_\alpha$ are étalé by proposition 3.5. The morphism $V_\alpha \hookrightarrow U$ is the inclusion of the affine open V_α. Write $V_{i,\alpha} := V_\alpha \times_U U_i$. Now by proposition 5.3 we can refine to an affine étalé cover

$$\{W_{i,\alpha,\lambda} \hookrightarrow V_{i,\alpha} \to V_\alpha\}_{i,\lambda}.$$

In particular $V_{i,\alpha} = \bigcup_\lambda W_{i,\alpha,\lambda}$. Now X, V_α and $W_{i,\alpha,\lambda}$ are affine. By lemma 3 sheaf condition holds for $\{W_{i,\alpha,\lambda} \to V_\alpha\}_{i,\lambda}$ for all α, i.e. the following diagram is a exact:

$$h_X(V_\alpha) \longrightarrow \prod_{(i,\lambda)} h_X(W_{i,\alpha,\lambda}) \rightrightarrows \prod_{(i,\lambda),(j,\mu)} h_X(W_{i,\alpha,\lambda} \times_{V_\alpha} W_{j,\alpha,\mu}) \tag{5.6}$$

We write $f_{i,\alpha,\lambda} : W_{i,\alpha,\lambda} \hookrightarrow V_{i,\alpha} \hookrightarrow U_i \xrightarrow{f_i} X$ for the restriction of f_i to $W_{i,\alpha,\lambda}$. Define $W := W_{i,\alpha,\lambda} \times_{V_\alpha} W_{j,\alpha,\mu}$.

We have canonical morphisms

$$\varphi : W \xrightarrow{\pi_1} W_{i,\alpha,\lambda} \hookrightarrow V_{i,\alpha} = V_\alpha \times_U U_i \xrightarrow{\pi_{U_i}} U_i$$
$$\psi : W \xrightarrow{\pi_1} W_{j,\alpha,\mu} \hookrightarrow V_{j,\alpha} = V_\alpha \times_U U_j \xrightarrow{\pi_{U_j}} U_j$$

We will now proof, that $f_i \circ \varphi = f_j \circ \psi$.
Since W is the fibre product over V_α, the following commute:

$$\begin{tikzcd}
& W \arrow[dl] \arrow[dr] & \\
W_{i,\alpha,\lambda} \arrow[d, hook] & & W_{j,\alpha,\mu} \arrow[d, hook] \\
V_{i,\alpha} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & V_\alpha \arrow[d] & V_{j,\alpha} \arrow[l] \arrow[d] \arrow[dl, phantom, "\llcorner", very near start] \\
U_i \arrow[r] & U & U_j \arrow[l]
\end{tikzcd}$$

By commutativity of 5.5, there is a unique δ fitting into the following diagram:

$$\begin{tikzcd}
& W \arrow[dl] \arrow[dr] \arrow[dd, dashed, "\delta"] & \\
W_{i,\alpha,\lambda} \arrow[d] & & W_{j,\alpha,\mu} \arrow[d] \\
V_{i,\alpha} = V_\alpha \times_U U_i \arrow[d] & U_i \times_U U_j \arrow[dl, "\pi_{U_i}"'] \arrow[dr, "\pi_{U_j}"] \arrow[d, phantom, "\vee", very near start] & V_{j,\alpha} = V_\alpha \times_U U_j \arrow[d] \\
U_i \arrow[r] \arrow[dr, "f_i"'] & U & U_j \arrow[l] \arrow[dl, "f_j"] \\
& X &
\end{tikzcd}$$

Hence $f_i \circ \pi_{U_i} = f_j \circ \pi_{U_j}$ pulls back to W via δ, which yields $f_{i,\alpha,\lambda} \mid_W = f_{j,\alpha,\mu} \mid_W$. By the sheaf condition 5.6, the $f_{i,\alpha,\lambda}$ glue together to a unique $g_\alpha : V_\alpha \to X$.

Next we prove, that the g_α agree on overlaps $V_{\alpha\beta} := V_\alpha \cap V_\beta \cong V_\alpha \times_U V_\beta$ and glue Zariski locally. Let $\{T_\eta\}_\eta$ be an affine open cover of $V_{\alpha\beta}$.

Again by proposition 3.5 the following pullback is étalé:

$$\begin{tikzcd} T_{i,\eta} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & V_{i,\alpha\beta} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & U_i \arrow[d] \\ T_\eta \arrow[r, hook] & V_{\alpha\beta} \arrow[r, hook] & U \end{tikzcd}$$

Again by proposition 5.3 we refine $T_{i,\eta}$ by an affine cover $\{T_{i,\eta,\lambda} \hookrightarrow T_{i,\eta} \to T_\eta\}_{i,\lambda}$. For each i, λ we can restrict f_i to $f_{i,\eta,\lambda} : T_{i,\eta,\lambda} \hookrightarrow T_{i,\eta} \to U_i \xrightarrow{f_i} X$. By construction g_α restrict uniquely to

The $V_{i,\alpha\beta}$ cover $V_{\alpha\beta}$, because U_i cover U. Since g_α restrict to $f_{i,\alpha,\lambda}$ on the affine open $W_{i,\alpha,\lambda}$, we get $g_\alpha |_{V_{i,\alpha\beta}} = f_i |_{V_{i,\alpha\beta}}$ for all i. Analogously $g_\beta |_{V_{i,\alpha\beta}} = f_i |_{V_{i,\alpha\beta}}$. So all g_α can be glued together to a unique morphism $f : U \to X$, since h_X is a sheaf in the Zariski site.

Now we assume X to be any scheme. Let $\{W_i \hookrightarrow X\}_{i\in I}$ be an open affine cover of the scheme X, and set

$$W_{ij} := W_i \cap W_j \qquad (i,j \in I).$$

Then via Zariksi gluing we have a natural isomorphim of schemes

$$X \cong \mathrm{Coeq}\Big(\coprod_{i,j} W_{ij} \rightrightarrows \coprod_i W_i\Big).$$

Then for every scheme S, by continuity of Hom there is a natural isomorphism

$$h_X(S) = \mathrm{Hom}_{\mathsf{Sch}}(S,X) \;\cong\; \mathrm{Eq}\left(\prod_i \mathrm{Hom}_{\mathsf{Sch}}(S,W_i) \rightrightarrows \prod_{i,j} \mathrm{Hom}_{\mathsf{Sch}}(S,W_{ij})\right),$$

where the two arrows are induced by restricting a morphism $S \to W_i$ to $S \to W_{ij}$ along the open immersions $W_{ij} \hookrightarrow W_i$ and $W_{ij} \hookrightarrow W_j$.

Via Yoneda, this is equivalent to an isomorphism of presheaves in $\mathsf{PSh}(\mathsf{Sch})$,

$$h_X \;\cong\; \mathrm{Eq}\left(\prod_i h_{W_i} \;\rightrightarrows\; \prod_{i,j} h_{W_{ij}}\right).$$

In particular, h_X is a limit of the diagram of the h_{W_i} and $h_{W_{ij}}$, hence also computed sectionwise for presheaves.

By the frist part of the proof, h_{W_i} and all $h_{W_{ij}}$ are sheaves. By corrolary 1, the category $\mathsf{Sh}(\mathsf{Sch}_{et})$ is closed under limits, hence h_X is a sheaf too. $\square$

6 Descent Theory

Proposition 6.1 (Descent for faithfully flat modules): *(***alper2026stacks***)* *Let $\varphi : A \to B$ be a faithfully flat ring map, then the following diagram is an equalizer:*

$$0 \longrightarrow A \xrightarrow{\varphi} B \underset{b \mapsto 1 \otimes b}{\overset{b \mapsto b \otimes 1}{\rightrightarrows}} B \otimes_A B$$

Let M be any A-module, then the following diagram is an equalizer:

$$0 \longrightarrow M \xrightarrow{m \mapsto m \otimes 1} M \otimes_A B \underset{m \otimes b \mapsto m \otimes 1 \otimes b}{\overset{m \otimes b \mapsto m \otimes b \otimes 1}{\rightrightarrows}} M \otimes_A B \otimes_A B$$

7 Algebraic Spaces

We have seen in ... that the representable presheaf of a scheme is indeeed a sheaf in $\mathsf{Sch}_{/Et}$. Via the Yoneda embedding we can identify a scheme U with its correspoding sheaf and denote it also by U.

Definition 7.1 (Representable Morphism): *(***alper2026stacks***) A morphism $X \to Y$ of sheaves over Sch is called representable by schemes, if for every scheme T and every morphism $T \to Y$, the fibre product $X \times_Y T$ is a scheme.*

Definition 7.2 (Algebraic Space): *(***alper2026stacks***) An algebraic space is a sheaf X on $\mathsf{Sch}_{/Et}$, such that there exists a scheme U and an étalé surjection $U \to X$ representable by schemes.*

In analogy with schemes, which are Zariski-locally given by affine schemes, an algebraic space is a sheaf admitting an étalé cover by schemes, and hence, after refinement (5.3), by affine schemes. Thus an algebraic space is a sheaf that is étalé-locally a scheme.

Indeed, let $U \to X$ be an étalé surjection, where U is a scheme, and let $\{U_i\}$ be an open affine cover of U. Then the induced morphism

$$\coprod_i U_i \to X$$

is again an étalé surjection. It follows that every algebraic space admits an étalé cover by affine schemes.

Theorem 3 (Representability of the Diagonal): *(**alper2026stacks**) The diagonal $\Delta_X : X \to X \times X$ of an algebraic space X is represantable by schemes.*

Lemma 4 (Quotiens of spaces by finite equivalence relation): *Let*

$$R \rightrightarrows U$$

be a finite equivalence relation in algebraic spaces. Then the sheaf quotient $X := U/R$ exists as an algebraic space.

Proof. First we prove it for the affine case. Assume $U = A$ $\square$

8 Algebraic Stacks

8.1 Fibred Categories

In this section we will consider commutative diagrams of the following form:

$$\begin{tikzcd} \zeta \arrow[r, "\phi"] \arrow[d, mapsto] & \eta \arrow[d, mapsto] \\ U \arrow[r, "f"] & V \end{tikzcd} \qquad \begin{tikzcd} \mathcal{F} \arrow[d, "p_{\mathcal{F}}"] \\ \mathcal{C} \end{tikzcd}$$

This denotes that ζ lies over U and ϕ lies over f, i.e. $p_{\mathcal{F}}(\zeta) = U$ and $p_{\mathcal{F}}(\phi) = f$.

Definition 8.1 (Category fibred in Groupoids): *(Gómez, 2001, p. 9, Definition 2.15) Let $\mathcal{F}$ be a category over $\mathcal{C}$. We call it a category fibred in groupoids, if*

1) Pullbacks exists, i.e. for all morphisms $f : B' \to B$ in $\mathcal{C}$ and every object X lying over B there exists an object X' and a morphism $\phi : X' \to X$ lying over B' and f resp.

$$\begin{tikzcd} X' \arrow[r, dashed, "\phi"] \arrow[d, dashed, mapsto] & X \arrow[d, mapsto] \\ B' \arrow[r, "f"'] & B \end{tikzcd}$$

2) Every diagram

$$\begin{tikzcd}
X_3 \arrow[rr, "\psi"] \arrow[dr, dashed, "\varphi"'] \arrow[dd, mapsto] & & X_1 \arrow[dd, mapsto] \\
& X_2 \arrow[ur, "\phi"'] \arrow[dd, mapsto] & \\
B_3 \arrow[rr, "f\circ f'"] \arrow[dr, "f'"'] & & B_1 \\
& B_2 \arrow[ur, "f"'] &
\end{tikzcd}$$

admits a unique morphism φ lying over f' and letting the upper triangle commute.

Definition 8.2 : *Let $\mathcal{F} \to \mathcal{C}$ be a category fibred in groupoids and let $B \in \mathcal{C}$. We call the subcategory of $\mathcal{F}$ of objects lying over B and morphisms lying over id_B the fibre of $\mathcal{F}$ over B and denote it by $\mathcal{F}(B)$.*

Proposition 8.3 : *The pullback X' of X along f is unique up to isomorphism. We will denote it by f^*X*

Proof. Assume there is a nother X'' satisfying 1). Apply 2) with $X_1 = X$ and $X_2 = X$” $X_3 = X'$ to optain a unique morphism $X' \to X''$. Then exchanging roles yields a unique morphism $X'' \to X'$ □

Proposition 8.4 : *In definition 8.1 2), f is an isomorphism if and only if ϕ is an isomorphism.*

Proof. One direction is obvious, since $p_{\mathcal{F}}$ is a functor. For the other direction consider the following diagram:

$$\begin{tikzcd}
X_1 \arrow[rr, "id"] \arrow[dr, dashed, "\phi^{-1}"'] \arrow[dd, mapsto] & & X_1 \arrow[dd, mapsto] \\
& X_2 \arrow[ur, "\phi"'] \arrow[dd, mapsto] & \\
B_1 \arrow[rr, "id"] \arrow[dr, "f^{-1}"'] & & B_1 \\
& B_2 \arrow[ur, "f"'] &
\end{tikzcd}$$

$\square$

Corollary 2 : *Let $\mathcal{F} \to \mathcal{C}$ be a category fibred in groupoids. Then the fibres $\mathcal{F}(B)$ of $\mathcal{F}$ over every object $B \in \mathcal{C}$ is a groupoid.*

Remark 3 : *As explained in the Stacks Project The Stacks Project Authors, 2024, Tags 02XJ, 003S, from a higher-categorical point of view, a category $\mathcal{F} \to \mathcal{C}$ fibred in groupoids may equivalently be regarded as a pseudofunctor $\mathcal{F} : \mathcal{C}^{\mathrm{op}} \to \mathsf{Grpd}$, where* Grpd *denotes the (2,1)-category of groupoids. This follows the Grothendieck construction, see The Stacks Project Authors, 2024, Tags 02XJ, 003S and Vistoli, 2007, p. 45, section 3.1.2.*

8.2 Stacks

Conceptually, stacks generalize sheaves by allowing values in groupoids instead of sets. Equivalently, stacks are 2-sheaves on a site, or sheaves valued in ∞-groupoids.

Definition 8.5 (Stack): *(The Stacks Project Authors, 2024, Tag 026F) Let $(\mathcal{C}, J)$ be a site, and let*

$$p_{\mathcal{F}} : \mathcal{F} \to \mathcal{C}$$

be a category fibred in groupoids. We call $\mathcal{F}$ a stack *over* $(\mathcal{C},J)$ *if for every object $B \in \mathcal{C}$ and every covering $\{f_i : U_i \to U\}_{i\in I}$ in J, the following conditions hold:*

1) ***Sheaf condition for isomorphisms.*** *For every pair of objects $X,Y \in \mathcal{F}(B)$, the presheaf*

$$Isom_U(X,Y) : (\mathcal{C}/U)^{op} \to \mathsf{Set}, \qquad (g : U' \to U) \longmapsto Isom_{\mathcal{F}(U')}(g^*X,g^*Y)$$

is a sheaf on the slice site $(\mathcal{C}/U,J_U)$.

2) ***Effective descent for objects.*** *Let $\{f_i : U_i \to U\}_{i\in I}$ be a covering, and for each i let $X_i \in \mathcal{F}(U_i)$ be an object. Suppose moreover that for every pair i,j there is an isomorphism*

$$\varphi_{ij} : \pi_1^*X_i \xrightarrow{\sim} \pi_2^*X_j \qquad \text{in } \mathcal{F}(U_i \times_U U_j),$$

where $\pi_1 : U_i \times_U U_j \to U_i$ and $\pi_2 : U_i \times_U U_j \to U_j$ are the projections, and that these isomorphisms satisfy the cocycle condition on triple fibre products

$$U_i \times_U U_j \times_U U_k.$$

Then there exists an object $X \in \mathcal{F}(U)$ together with isomorphisms

$$\alpha_i : f_i^*X \xrightarrow{\sim} X_i \qquad \text{in } \mathcal{F}(U_i)$$

such that for every pair i,j the diagram

$$\begin{tikzcd} \pi_1^*f_i^*X \arrow[r, "\pi_1^*\alpha_i"] \arrow[d, "\cong"'] & \pi_1^*X_i \arrow[d, "\varphi_{ij}"] \\ \pi_2^*f_j^*X \arrow[r, "\pi_2^*\alpha_j"'] & \pi_2^*X_j \end{tikzcd}$$

commutes in $\mathcal{F}(U_i \times_U U_j)$.

Remark 4 : *Following Definition 2.6 of Khan, 2022, p. 13 we get a nice ∞-categorical picture of a stack, namely it is a 2-sheaf, satisfying descent*

$$\mathcal{X}(U) \longrightarrow \prod \mathcal{X}(U_i) \rightrightarrows \prod_{i,j} \mathcal{X}(U_i \times_U U_j) \substack{\rightarrow\\\rightarrow\\\rightarrow} \prod_{i,j,k} \mathcal{X}(U_i \times_U U_j \times_U U_k)$$

Definition 8.6 (Algebraic Stack): *(The Stacks Project Authors, 2024, Tag 026O) Let S be a scheme. An algebraic stack over S is a category fibred in groupoids*

$$\mathcal{X} \to (\mathsf{Sch}_{/S})_{\mathrm{fppf}}$$

such that:

1) *$\mathcal{X}$ is a stack in groupoids over $(\mathsf{Sch}_{/S})_{\mathrm{fppf}}$;*

2) *the diagonal*

$$\Delta_{\mathcal{X}} \colon \mathcal{X} \to \mathcal{X} \times \mathcal{X}$$

is representable by algebraic spaces;

3) *there exists a scheme U and a smooth surjective morphism*

$$U \to \mathcal{X}.$$

Proposition 8.7 : *There is a fully faithful embedding of algebraic spaces into algebraic stacks $AlgSp \subset AlgStacks$.*

Definition 8.8 (Inertia stack): *(The Stacks Project Authors, 2024, Tag 036X) Let $\mathcal{X}$ be an algebraic stack over S. The inertia stack of $\mathcal{X}$, denoted*

$$I_{\mathcal{X}},$$

is the stack over $(\mathsf{Sch}_{/S})_{\mathrm{fppf}}$ defined as follows:

For every scheme U over S, the groupoid $I_{\mathcal{X}}(U)$ has

1) objects given by pairs (x,α)*, where*

$$x \in \mathcal{X}(U) \quad \textit{and} \quad \alpha \in \mathrm{Aut}_{\mathcal{X}(U)}(x),$$

2) morphisms

$$(x,\alpha) \to (y,\beta)$$

given by morphisms $f : x \to y$ *in* $\mathcal{X}(U)$ *such that*

$$f \circ \alpha = \beta \circ f.$$

Lemma 5 : *Let* $\mathcal{X}$ *be an algebraic stack over* S*. Then there is an isomorphism of stacks*

$$I_{\mathcal{X}} \cong \mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X},$$

where both morphisms $\mathcal{X} \to \mathcal{X} \times \mathcal{X}$ *are equal to the diagonal*

$$\Delta_{\mathcal{X}} : \mathcal{X} \to \mathcal{X} \times \mathcal{X}.$$

Proof. Let U be a scheme over S. By definition of the fibre product of stacks, an object of

$$(\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X})(U)$$

consists of objects $x,y \in \mathcal{X}(U)$ together with an isomorphism

$$\varphi : \Delta_{\mathcal{X}}(x) \xrightarrow{\sim} \Delta_{\mathcal{X}}(y)$$

in $(\mathcal{X} \times \mathcal{X})(U)$.

Now

$$\Delta_{\mathcal{X}}(x) = (x,x) \quad \text{and} \quad \Delta_{\mathcal{X}}(y) = (y,y).$$

Hence an isomorphism

$$\varphi : (x,x) \xrightarrow{\sim} (y,y)$$

is given by a pair of isomorphisms

$$(f,g)$$

with

$$f : x \xrightarrow{\sim} y \quad \text{and} \quad g : x \xrightarrow{\sim} y$$

in $\mathcal{X}(U)$.

From such a tuple (x,y,f,g) we obtain a pair

$$(x,\alpha), \qquad \alpha = g^{-1} \circ f \in \mathrm{Aut}_{\mathcal{X}(U)}(x).$$

Conversely, given a pair (x,α) with

$$\alpha \in \mathrm{Aut}_{\mathcal{X}(U)}(x),$$

we obtain an object of the fibre product by taking

$$x = y$$

and the isomorphism

$$(\alpha,id_x) : (x,x) \xrightarrow{\sim} (x,x).$$

These two constructions are inverse to each other up to canonical isomorphism. Thus the objects of

$$(\mathcal{X} \times_{\mathcal{X}\times\mathcal{X}} \mathcal{X})\,(U)$$

identify with pairs (x,α), where $x \in \mathcal{X}(U)$ and

$$\alpha \in \mathrm{Aut}_{\mathcal{X}(U)}(x).$$

It remains to identify the morphisms. A morphism in the fibre product from

$$(x,y,f,g) \quad \text{to} \quad (x',y',f',g')$$

is given by morphisms

$$a : x \to x', \qquad b : y \to y'$$

in $\mathcal{X}(U)$ such that the square

$$\begin{tikzcd}
(x,x) \arrow[r, "{(f,g)}"] \arrow[d, "{(a,a)}"'] & (y,y) \arrow[d, "{(b,b)}"] \\
(x',x') \arrow[r, "{(f',g')}"] & (y',y')
\end{tikzcd}$$

commutes in $(\mathcal{X} \times \mathcal{X})(U)$. This means precisely that

$$b \circ f = f' \circ a \qquad \text{and} \qquad b \circ g = g' \circ a.$$

Passing to the corresponding automorphisms

$$\alpha = g^{-1} \circ f \qquad \text{and} \qquad \beta = (g')^{-1} \circ f',$$

one checks that this is equivalent to

$$a \circ \alpha = \beta \circ a.$$

Hence morphisms correspond exactly to morphisms

$$a : x \to x'$$

such that

$$a \circ \alpha = \beta \circ a.$$

Therefore the stack

$$\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X}$$

is identified with the inertia stack $I_{\mathcal{X}}$. □

Corollary 3 : *There is a natural morphism*

$$I_{\mathcal{X}} \to \mathcal{X}$$

given by either projection from the fibre product. On objects it sends

$$(x,\alpha) \mapsto x.$$

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Appendix A

Code Examples

A.1 AMPL

```
reset;  # all blue words are keywords in this language
        # they are defined in the file 01_Document_administration/
            f_CodeLanguageSpecifications.tex

model simulation.mod;
data simulation.dat;
include initial.dat;
option ipoptoptions "halt_on_ampl_error␣yes";

let e0_param := 4;

solve;
```

Code A.1: Code example for AMPL.

A.2 Matlab

```
clc
clear
close all

e0_param = 4;

j=0;
```

```
for i=1:e0_params % this loop is incredibly smart
    if 1==2
        j=j+1;
    else
        j=j-1;
    end
end
```

Code A.2: Code example for Matlab.

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